SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY
MUKKA,MANGALURU - 574146

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING ENGINEERING

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
GROUP TASK REPORT
ON

"DATA FOR AI PROJECT SIMULATION, ML ETHICS AND BIAS CASE STUDY, BACKPROPOGATION WALKTHROUGH, CONTENT AI EVALUATION CHALLENGE "

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Submitted By,

KRISHNAVENI	**01SU25AI036**
NANDANA A S	**01SU25AI047**
NANDITA NAIK	**01SU25AI048**
POOJA	**01SU25AI053**

UNDER THE GUIDANCE OF

PROF. MAHESH KUMAR VB

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING ENGINEERING

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MODULE 2 — Data for AI Project Simulation

2.1 Introduction and Problem Statement

Mock project: AI-Based Recommendation System for Students. Students face many course, project and resource choices, and generic lists ignore individual goals. The proposed system suggests learning resources by matching each student's declared interests and completed activities to resource features. This is a *hypothetical* simulation; no real student data is used.

2.2 Project Objectives and Scope

The objectives are to (i) recommend relevant resources, (ii) collect only necessary data with consent, (iii) explain every recommendation and (iv) evaluate quality and fairness. The scope covers one department, optional participation and no automated decisions about grades or admission.

2.3 Proposed System Architecture and Workflow

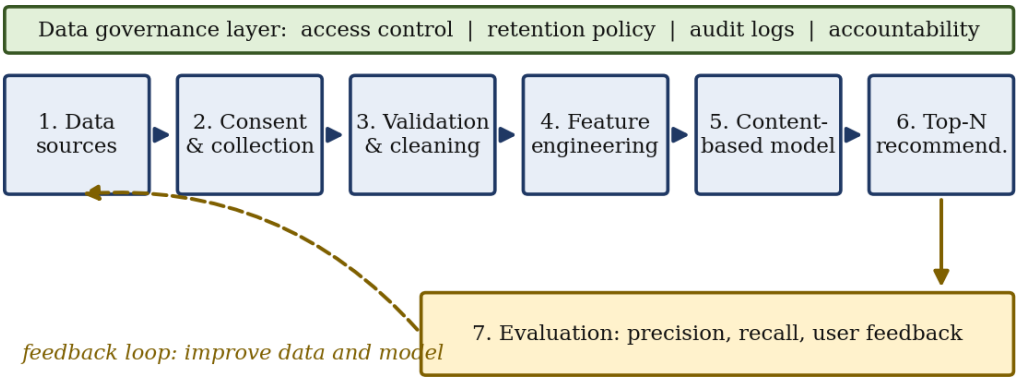


Figure 2.1: Project workflow and governance layer (hypothetical design)

2.4 Data Requirements, Sources and Types

Table 2.1: Data types in the project (illustrative classification)

Type	Example in this project	Handling
Structured	Course ID, semester, rating (1–5), skill tags	Stored in tables; validated against allowed values [3].
Unstructured	Free-text feedback, resource descriptions	Text processing; removal of personal details first [3].
Sensitive	USN, name, e-mail, disability or accessibility needs	Collected only if essential; encrypted, restricted access [1].
Big data	Not required for a department pilot	Would apply only if logs of several institutions were combined [3].

2.5 Data Collection and Consent

Participation is voluntary. Students receive a plain-language notice stating what is collected, why, who can see it, how long it is kept and how to withdraw. This supports data minimization, transparency and purpose limitation [1], [6], [7]. India's Digital Personal Data Protection Act, 2023 [6] and the EU GDPR [7] are consent-based, purpose-bound regimes and give special attention to children's data; a real deployment needs legal review.

2.6 Data Preparation

Table 2.2: Data preparation steps [Ext: standard practice; 3, 8]

Step	Action in the project	Example
Validation	Check types, ranges and allowed values	Rating must be 1–5
Missing values	Drop rows missing key IDs; impute rating with the median	Blank rating → 3
Duplicates	Remove repeated (student, resource) rows	Same click logged twice
Cleaning / normalization	Standardise text case; scale numeric features to 0–1	(x − min)/(max − min)
Encoding	One-hot encode categories such as level and topic	Level = Beginner → [1,0,0]
Feature selection	Keep only features that serve the recommendation purpose; drop identifiers	Drop name and e-mail
Train/test split	Hold out about 20% of interactions per student for testing	80/20 split [8]

2.7 Content-Based Recommendation Algorithm and Example

Content-based filtering recommends items whose features resemble items the user liked [2], [3]. Each resource is a binary vector over five features: Python, Statistics, Web, Databases, Algorithms. The student profile is the mean vector of liked resources, and resources are ranked by cosine similarity: $\cos(p, c) = (p \cdot c) / (\|p\| \, \|c\|)$.

Table 2.3: Illustrative recommendation example (invented data)

Candidate resource	Vector (Py, St, Web, DB, Alg)	p · c	‖c‖	Cosine
C1 Machine Learning Basics	(1, 1, 0, 0, 0)	1.00	1.414	**0.8165**
C4 Data Analytics with Python	(1, 1, 0, 1, 0)	1.00	1.732	**0.6667**
C3 Data Structures	(0, 0, 0, 1, 1)	0.50	1.414	0.4082
C2 Web Development	(1, 0, 1, 1, 0)	0.50	1.732	0.3333

The student liked "Python Programming" (1,0,0,0,1) and "Probability and Statistics" (0,1,0,0,0), so the profile is p = (0.5, 0.5, 0, 0, 0.5) with ‖p‖ = 0.866. C1 and C4 are recommended first, with an explanation such as "matches your interest in Python and Statistics".

2.8 Evaluation Methods

Precision@K is the fraction of the K recommended items that are relevant; recall is the fraction of all relevant items that were recommended [2], [9]. *Illustration:* if K = 3 and two of the three recommendations are later judged relevant out of three relevant items in total, precision = 2/3 ≈ 0.67 and recall = 2/3 ≈ 0.67. Short user surveys on usefulness and clarity of explanations complement these metrics, and results should be compared across student groups to check fairness.

2.9 Ethical Concerns and Data Governance

Table 2.4: Ethical concerns and governance controls ([PDF] terminology, applied by the author)

Principle	Concern	Control
Privacy	Tracking learning behaviour reveals personal habits	Data minimization; pseudonymous IDs; retention limit (e.g., one semester) [1]
Security	Leaks of student records	Encryption; role-based access control; audit logs [1]
Transparency	Opaque suggestions	Show "why recommended"; publish a data sheet [5]
Fairness	Popular topics crowd out niche interests	Diversity re-ranking; group-wise metrics [1]
Accountability	No owner for errors	Named data steward; complaint and opt-out route [1]
Accessibility	Interface unusable for some students	Screen-reader support; plain language

Governance: access is granted by role (student, faculty, administrator); data are kept for a stated period and then deleted; every access and model change is logged; and a named person is accountable for reviews [1], [5].

2.10 Expected Results, Limitations and Future Work

A pilot would be expected to give relevant, explainable suggestions for students with some history. Limitations include the cold-start problem (new students without history), narrow "filter-bubble" recommendations and small samples [2]. Future work includes hybrid filtering, feedback-based re-ranking and fairness audits across departments.

2.11 References

[1] *Data Classification Game*, uploaded course activity .
[2] F. Ricci, L. Rokach and B. Shapira, Eds., *Recommender Systems Handbook*, 3rd ed. Springer, 2022. doi:10.1007/978-1-0716-2197-4.
[3] J. Han, M. Kamber and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2012.
[4] C. C. Aggarwal, *Recommender Systems: The Textbook*. Springer, 2016. doi:10.1007/978-3-319-29659-3.
[5] T. Gebru *et al.*, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021. doi:10.1145/3458723.
[6] Government of India, *The Digital Personal Data Protection Act, 2023* (Act No. 22 of 2023).

MODULE 3 — ML Ethics and Bias Case Study

3.1 Introduction to Machine-Learning Ethics

Machine-learning ethics asks whether systems that learn from data treat people fairly, protect their rights, can be explained and have clear human accountability [2], [6]. Bias can enter through data, labels, features, model design and use [3]. This module analyses the experimental AI recruitment tool at Amazon, as reported by Reuters in 2018 [1]. Statements tagged [Reported] come from that article; statements tagged [Analysis] are the author's own reasoning and recommendations.

3.2 Background, Purpose and Reported Problems

Table 3.1: What the original reporting says [1]

Topic	[Reported] — paraphrased from Dastin, Reuters, 10 Oct 2018
Purpose	A team had been building programs since 2014 to review resumes and mechanise the search for top talent; candidates were scored from one to five stars.
Training data	The models learned from patterns in resumes submitted to the company over a ten-year period, most of which came from men, reflecting male dominance in the tech industry.
Gender bias found	By 2015 the company realised the system was not rating software-developer and other technical candidates in a gender-neutral way. It penalised resumes containing the word "women's" (for example "women's chess club captain") and downgraded graduates of two all-women's colleges.
Other signals	The system reportedly favoured verbs more common on male engineers' resumes and gave little weight to skills common across IT applicants. Gender bias was not the only issue: data problems also led to poorly matched candidates being recommended.
Response	Amazon edited the programs to be neutral to the specific terms found, but the sources said this gave no guarantee that the system would not find other discriminatory ways of sorting candidates. The project was later abandoned.
Use and comment	Sources said recruiters looked at the tool's recommendations but never relied solely on them. Amazon said the tool was never used by recruiters to evaluate candidates and stated a commitment to workforce diversity.

3.3 How Data, Labels, Features and Proxies Produce Bias [Analysis]

If past hiring favoured men, a model trained to imitate past "successful" hires learns that pattern as if it were merit [2]. Labels. The label "was hired" records past human decisions, not true job performance, so it carries earlier prejudice. Features and proxies. For this reason, removing sensitive attributes alone does not eliminate bias, and patching individual words (as reported) cannot guarantee fairness.

3.4 Effects and Causes

Table 3.2: Potential effects and causes [Analysis]

Dimension	Potential effect	Technical / organisational cause
Applicants	Qualified women ranked lower; lost opportunity; no explanation	Unrepresentative data; opaque scoring
Organisation	Less diverse talent; legal and reputational risk; wasted effort	No fairness goal set at the start; weak review
Technical	Pattern amplification; proxy leakage; accuracy hides group gaps	Optimising overall accuracy only; no subgroup testing

3.5 Proposed Solutions [Analysis]

Table 3.3: Mitigation measures

Measure	How it helps
Data documentation	Datasheets and model cards record data source, time period, known gaps and intended use [5], [6].
Representative datasets	Collect or re-weight data so relevant groups are adequately represented; review whether past labels are a fair target [3].
Fairness testing	Compare selection rates and error rates across groups before and after deployment.
Suitable metrics	Selection-rate ratio (the "four-fifths" rule of thumb in US employment guidelines [7]), equal opportunity [4]. Several fairness criteria cannot all hold at once, so the choice must be justified [8].
Human review and appeal	A trained person makes the final decision; applicants can request review.
Independent audit	External reviewers check data, code, metrics and outcomes against stated goals [9].
Continuous monitoring	Track group-wise outcomes after launch; retrain or retire the system when drift or harm appears [9].

3.6 Practical Fairness Audit Checklist

Table 3.4: Checklist for a hiring-model audit (tick during review)

Tick	Question
☐	Are the label and data period examined for past discrimination?
☐	Have proxy features (names, clubs, colleges, wording) been tested?
☐	Are accuracy, selection rate and error rates reported for each group?
☐	Is there a human decision-maker, an appeal route and an explanation for applicants?
☐	Are monitoring, audit dates and an accountable owner defined?

3.7 Limitations and Lessons Learned

Limitations: the case rests on anonymous sources and Amazon's partial comments, and the proposals above are general practice, not Amazon's actual remedies. Lessons: define fairness *before* building; historical data are not neutral; patching symptoms is insufficient; oversight, documentation and auditing are part of the system.

3.8 Conclusion and Learning Outcomes

- Explain how data, labels and proxies create bias, and why dropping a sensitive attribute is insufficient; automating a past process can automate its injustice.
- Separate documented facts from analysis; apply an audit checklist and fairness metrics.

3.9 References

[1] J. Dastin, "Amazon scraps secret AI recruiting tool that showed bias against women," *Reuters*, 10 Oct. 2018. https://www.reuters.com/article/us-amazon-com-jobs-automation-insight/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08G.

[2] S. Barocas, M. Hardt and A. Narayanan, *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, 2023. https://fairmlbook.org.

[3] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Computing Surveys*, vol. 54, no. 6, art. 115, 2021. doi:10.1145/3457607.

[4] M. Hardt, E. Price and N. Srebro, "Equality of opportunity in supervised learning," *NeurIPS*, 2016. arXiv:1610.02413.

[5] M. Mitchell *et al.*, "Model cards for model reporting," *Proc. ACM Conf. Fairness, Accountability, and Transparency*, pp. 220–229, 2019. doi:10.1145/3287560.3287596.

[6] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, 2023. doi:10.6028/NIST.AI.100-1.

[7] US Equal Employment Opportunity Commission *et al.*, Uniform Guidelines on Employee Selection Procedures, 29 C.F.R. Part 1607, 1978.

[8] A. Chouldechova, "Fair prediction with disparate impact," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017. doi:10.1089/big.2016.0047.

[9] R. Schwartz *et al.*, *Towards a Standard for Identifying and Managing Bias in Artificial Intelligence*, NIST SP 1270, 2022. doi:10.6028/NIST.SP.1270.

MODULE 4 — Backpropagation Walkthrough

4.1 Introduction to Artificial Neural Networks

An artificial neural network is a set of connected units (neurons) arranged in layers. Each neuron computes a weighted sum of its inputs plus a bias and passes it through an activation function [2], [3]. Learning means adjusting weights and biases to reduce a loss. Backpropagation applies the chain rule of calculus to compute how the loss changes with respect to every parameter, and gradient descent then moves each parameter a small step against its gradient [1], [2]. The example below uses a 2–2–1 network, sigmoid activations, one training example, mean squared error and learning rate $\eta = 0.5$.

4.2 Architecture, Notation and Initial Values

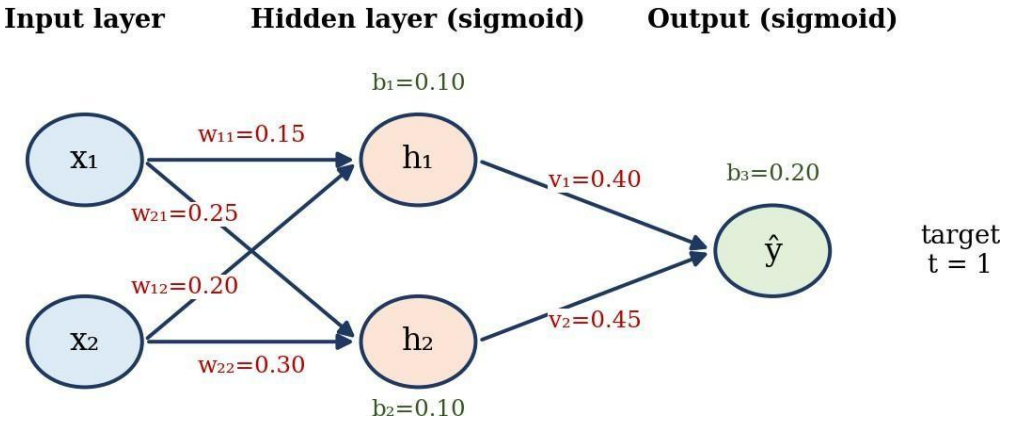


Figure 4.1: 2–2–1 network with initial weights and biases

Symbol	Meaning (simple English)
x_1, x_2; t	Inputs and the target output. Here $x_1 = 0.5$, $x_2 = 1.0$ and $t = 1$.
w_{ij}, b_i	Weight from input j to hidden neuron i, and the bias of hidden neuron i. v_i and b_3 are the output weights and bias.
z, h, $\hat{y}$	z is the weighted sum; $h = \sigma(z)$ is a hidden activation; $\hat{y} = \sigma(z_o)$ is the network's prediction.
$\sigma(z) = 1/(1 + e^{-z})$	Sigmoid activation. Its slope is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
$L = \frac{1}{2}(t - \hat{y})^2$	Squared error loss for one example (the ½ simplifies the derivative).
δ	Error signal of a neuron: $\delta = \partial L/\partial z$. $\eta = 0.5$ is the learning rate.

4.3 Forward Propagation

Table 4.1: Forward pass (values rounded to 4 decimals)

Quantity	Equation and substitution	Result
Hidden sum z_1	$w_{11}x_1 + w_{12}x_2 + b_1 = 0.15(0.5) + 0.20(1.0) + 0.10 = 0.0750 + 0.2000 + 0.1000$	0.3750
Activation h_1	$\sigma(0.3750) = 1/(1 + e^{-0.3750})$	0.5927
Hidden sum z_2	$w_{21}x_1 + w_{22}x_2 + b_2 = 0.25(0.5) + 0.30(1.0) + 0.10 = 0.1250 + 0.3000 + 0.1000$	0.5250

Quantity	Equation and substitution	Result
Activation h_2	$\sigma(0.5250)$	0.6283
Output sum z_o	$v_1h_1 + v_2h_2 + b_3 = 0.40(0.5927) + 0.45(0.6283) + 0.20 = 0.2371 + 0.2827 + 0.2000$	0.7198
Prediction $\hat{y}$	$\sigma(0.7198)$	0.6726
Loss L	$\frac{1}{2}(t - \hat{y})^2 = \frac{1}{2}(1 - 0.6726)^2 = \frac{1}{2}(0.3274)^2 = \frac{1}{2}(0.1072)$	**0.0536**

4.4 Backpropagation Using the Chain Rule

The chain rule links the loss to a weight through the quantities in between. For an output weight: $\partial L/\partial v_1 = (\partial L/\partial \hat{y})(\partial \hat{y}/\partial z_o)(\partial z_o/\partial v_1)$. Here $\partial L/\partial \hat{y} = -(t - \hat{y}) = \hat{y} - t$, $\partial \hat{y}/\partial z_o = \hat{y}(1 - \hat{y})$ and $\partial z_o/\partial v_1 = h_1$. Writing $\delta_o = (\hat{y} - t)\cdot\hat{y}(1 - \hat{y})$ gives $\partial L/\partial v_1 = \delta_o h_1$. For a hidden neuron the error signal is passed back through the output weight: $\delta_i = \delta_o \cdot v_i \cdot h_i(1 - h_i)$, and $\partial L/\partial w_{ij} = \delta_i x_j$.

Table 4.2: Backward pass (gradients; negative sign means "increase this parameter to reduce loss")

Quantity	Equation and substitution	Result
Output delta δ_o	$(\hat{y} - t)\cdot\hat{y}(1 - \hat{y}) = (0.6726 - 1)(0.6726)(0.3274) = (-0.3274)(0.2202)$	−0.0721
$\partial L/\partial v_1$	$\delta_o h_1 = (-0.0721)(0.5927)$	−0.0427
$\partial L/\partial v_2$	$\delta_o h_2 = (-0.0721)(0.6283)$	−0.0453
$\partial L/\partial b_3$	δ_o	−0.0721
Hidden delta δ_1	$\delta_o v_1 h_1(1 - h_1) = (-0.0721)(0.40)(0.5927 \times 0.4073) = (-0.02884)(0.2414)$	−0.0070
Hidden delta δ_2	$\delta_o v_2 h_2(1 - h_2) = (-0.0721)(0.45)(0.6283 \times 0.3717) = (-0.03245)(0.2336)$	−0.0076
$\partial L/\partial w_{11}$, $\partial L/\partial w_{12}$, $\partial L/\partial b_1$	$\delta_1 x_1$, $\delta_1 x_2$, $\delta_1 = (-0.0070)(0.5)$, $(-0.0070)(1.0)$, -0.0070	−0.0035, −0.0070, −0.0070
$\partial L/\partial w_{21}$, $\partial L/\partial w_{22}$, $\partial L/\partial b_2$	$\delta_2 x_1$, $\delta_2 x_2$, $\delta_2 = (-0.0076)(0.5)$, $(-0.0076)(1.0)$, -0.0076	−0.0038, −0.0076, −0.0076

4.5 Gradient Descent Update and New Loss

Each parameter is updated by $\theta \leftarrow \theta - \eta \cdot \partial L/\partial \theta$ with $\eta = 0.5$. All gradients above were computed with the **old** output weights, and all updates are applied afterwards. Because every gradient is negative, every parameter increases.

Table 4.3: Parameter updates ($\eta = 0.5$)

Weights	Update	New value	Related bias (update → new)
v_1 / v_2	$0.40 - 0.5(-0.0427)$; $0.45 - 0.5(-0.0453)$	0.4214 / 0.4727	b_3: $0.20 - 0.5(-0.0721) = 0.2361$
w_{11} / w_{12}	$0.15 - 0.5(-0.0035)$; $0.20 - 0.5(-0.0070)$	0.1517 / 0.2035	b_1: $0.10 - 0.5(-0.0070) = 0.1035$
w_{21} / w_{22}	$0.25 - 0.5(-0.0038)$; $0.30 - 0.5(-0.0076)$	0.2519 / 0.3038	b_2: $0.10 - 0.5(-0.0076) = 0.1038$

Check. Repeating the forward pass with the new values gives $z_1 = 0.3828$, $h_1 = 0.5946$, $z_2 = 0.5335$, $h_2 = 0.6303$, $z_o = 0.7845$ and $\hat{y} = 0.6866$. The loss falls from **0.0536 to 0.0491** (about 8%), and the prediction moves toward $t = 1$, so the gradient signs are correct. The values were

also checked against a numerical finite-difference gradient. *Rounded values may cause small differences (about ±0.0001) if you recompute by hand.*

4.6 Flowchart and Common Calculation Errors

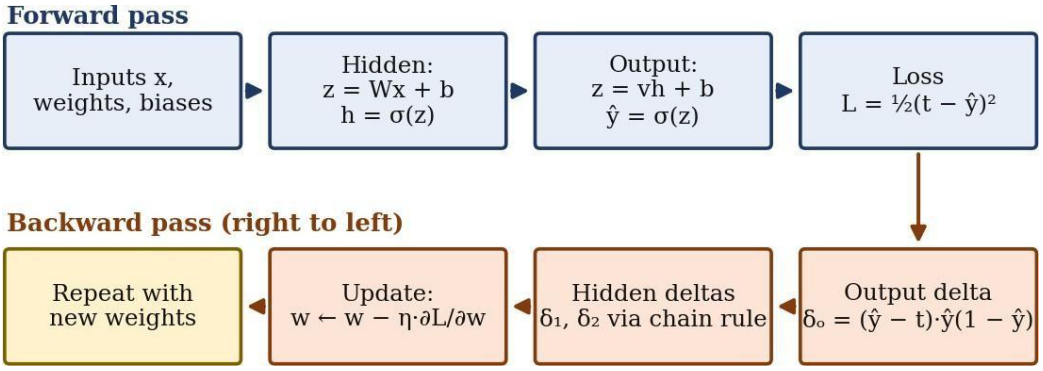


Figure 4.2: Forward and backward propagation flow

Table 4.4: Common errors

Error	Correction
Forgetting the sigmoid slope $\sigma' = \sigma(1-\sigma)$	Multiply by $\hat{y}(1-\hat{y})$ and $h_i(1-h_i)$ at every neuron.
Wrong sign (using $t-\hat{y}$ instead of $\hat{y}-t$)	With $L = \frac{1}{2}(t-\hat{y})^2$, $\partial L/\partial \hat{y} = \hat{y} - t$.
Using updated v_1, v_2 in the hidden gradients	Use the old weights; update everything after all gradients are found.
Rounding too early; mixing up indices	Keep 6 decimals in intermediate steps; check w_{ij} labels against the diagram.

4.7 Applications and Learning Outcomes

Backpropagation trains networks used in image recognition, speech and language processing, medical diagnosis and forecasting [4]. After this module the student should be able to perform a forward pass, apply the chain rule to obtain gradients, update weights with gradient descent, and verify that the loss decreased.

4.8 References

[1] D. E. Rumelhart, G. E. Hinton and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, 1986. doi:10.1038/323533a0.

[2] I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*. MIT Press, 2016, ch. 6. https://www.deeplearningbook.org.

[3] M. A. Nielsen, *Neural Networks and Deep Learning*. Determination Press, 2015, ch. 2. http://neuralnetworksanddeeplearning.com.

[4] Y. LeCun, Y. Bengio and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015. doi:10.1038/nature14539.

MODULE 5 — Content AI Evaluation Challenge

5.1 Introduction, Objectives and Tools

Natural Language Processing (NLP) enables computers to analyse and generate human language, and modern generative AI assistants are built on large neural language models [1], [2], [3]. Such tools write fluently but can state errors confidently, so learners must evaluate their output. Objectives: generate educational content with an AI assistant, evaluate it against defined criteria, improve it by human editing and reflect on ethics. Tools: a generative assistant (for example ChatGPT, Claude, Gemini or Microsoft Copilot), a grammar checker (for example Language Tool or the Word editor) and a readability measure such as the Flesch Reading Ease score [7].

5.2 Topic

Topic: "Overfitting in machine learning" for first-year BTech students. Prompt given to the tool: *"Write a 120-word explanation of overfitting in machine learning for first-year engineering students. Use simple English, one everyday analogy and one practical tip. Avoid jargon and do not invent statistics."*

5.3 Illustrative AI-Generated Sample and Human-Edited Version

The following texts are illustrative and were written for this exercise; they are not recorded outputs of a specific tool. A deliberate error was included in the draft so that evaluation can be practised.

Illustrative AI draft	Human-edited version
Overfitting happens when a model is too simple and cannot learn the pattern in the data. It is like a student who studies only a few topics. Studies show 90% of beginner models overfit. To avoid it, always use the largest possible model.	Overfitting happens when a model learns the training data too closely, including its noise, and then performs poorly on new data. It is like a student who memorises last year's answers but cannot solve a new question. A practical tip is to test the model on separate data and use simpler models or more training data when the gap is large.

5.4 Evaluation of the Text

Table 5.1: Text evaluation of the illustrative draft (scores 1–5 is illustrative)

Criterion	Observation	Score
Clarity and readability	Short sentences, easy to read; Flesch score can be computed to confirm.	4
Tone and audience	Friendly and suitable for beginners.	4
Factual accuracy	Wrong definition (this describes underfitting); unsupported "90%" claim; harmful advice to use the largest model.	1
Bias and inclusiveness	Neutral; analogy assumes a student context only.	4
Relevance	Addresses the topic, but the analogy does not match the definition.	3
Grammar and language	Correct and fluent.	5

5.5 Multimedia Presentation Evaluation

If the content is presented as a narrated slide video, the following rubric is applied. No recording is available for this report, so no scores or observations about voice, face or body language are claimed; the score column is left blank to be completed from an actual recording.

Table 5.2: Rubric for the presentation (1 = poor, 5 = excellent)

Criterion	What to check	Score (1–5)
Voice clarity and pronunciation	Clear, audible, technical terms pronounced correctly	[ ]
Pace and pauses	Comfortable speed; pauses before key points	[ ]
Facial expression and eye contact	Engaged expression; looks at the camera or audience	[ ]
Posture and gestures	Upright, purposeful, not distracting	[ ]
Speech–slide synchronisation	Speech matches the slide on screen; slides not overloaded	[ ]

5.6 Comparison, Ethical Issues and Improvements

The human-edited version corrects the definition, removes the invented statistic, aligns the analogy with the concept and adds an accurate tip, raising factual accuracy from 1 to 5 in this illustration. Ethical issues include hallucinations; privacy; copyright and attribution; transparency. Improvements: ask the tool for sources and verify them independently, give a more detailed prompt, check against a textbook, edit the draft and record what was changed.

5.7 References

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